

Lick Kast Imaging Polarimetry — Exposure Time & S/N Plan

UV-excess QSO candidates (34 targets with g photometry)

Methodology (K. Leighly notes 2026-07; E. Glikman's synphot notebook; Chromey, "To Measure the Sky", sec. 9.5.4)

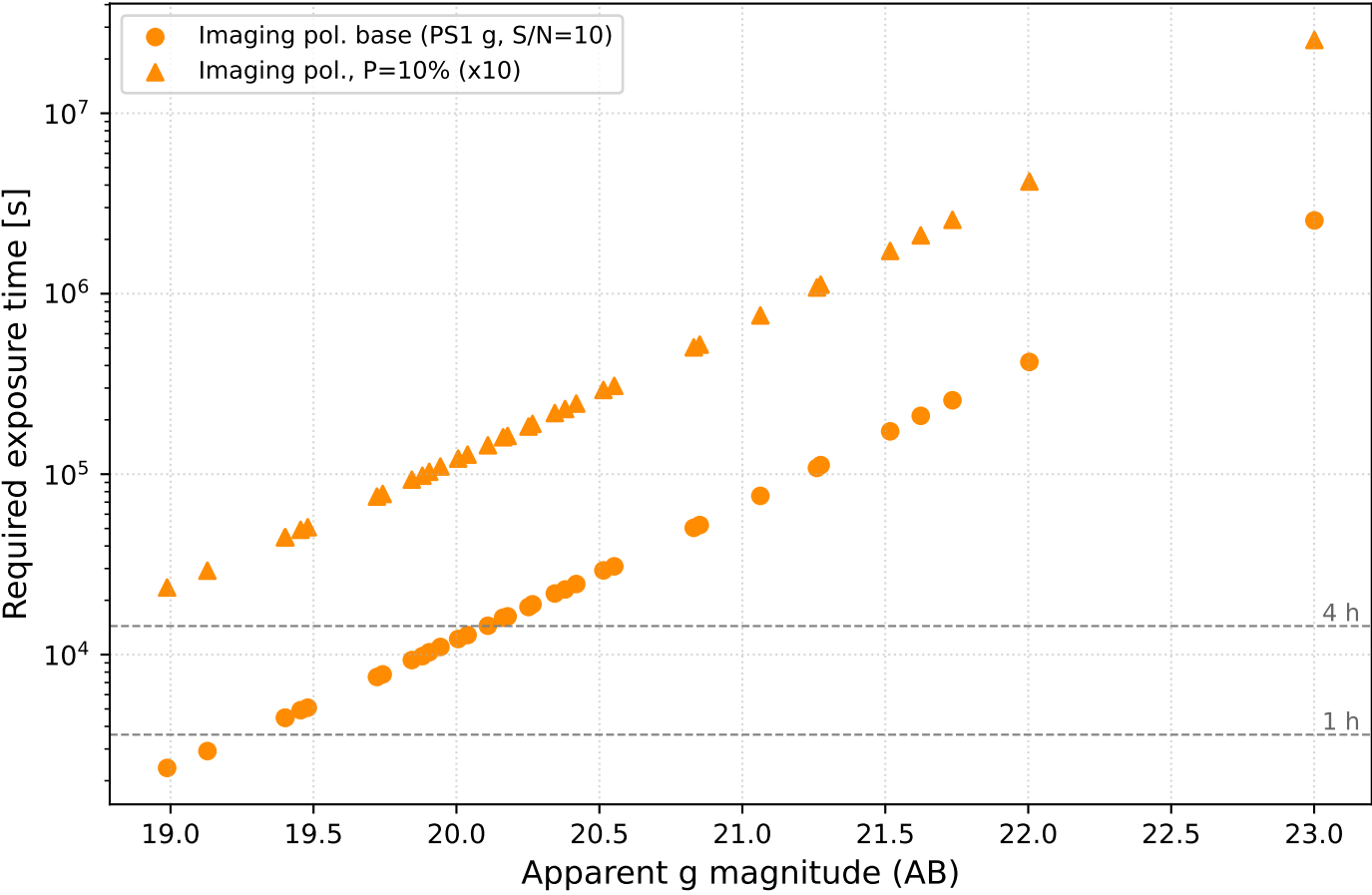
1. Each target is run through the Kast ETC (https://etc.ucolick.org/web_s2n/kast) with a flat template and its PS1 g-band AB magnitude + 0.752 mag: the polarization optics pass half the flux ($m_{\text{pol}} = m + 2.5 \log 2$).
2. ETC settings, identical for all targets: dichroic d55, slit 2.0 arcsec, binning 1x1 pixels, grism 600/4310, grating 600/7500, seeing 1.5", airmass 1.1, reference exposure 900 s.
3. The ETC's per-pixel object/sky/noise counts vs. wavelength are integrated over the PS1 g bandpass via synphot's `Observation.effstim()` (a throughput-weighted mean), then combined with the CCD equation $S/N = N_{\text{qso}} / \sqrt{N_{\text{qso}} + N_{\text{sky}} + \text{Noise}}$.
4. The exposure time reaching the required S/N is the exact solution of the CCD equation (Chromey eq. 9.77). Required S/N = 10 ; the 5 floor ("definitely >5") is plotted and tabulated.
5. Polarimetry penalty: to be sensitive to polarization fraction P, the normal exposure time is multiplied by 1/P (rough rule per notes) – x10 for P = 10%, x100 for P = 1%. Note P is NOT the scattered fraction: scattering with unfavorable geometry can cancel to zero net polarization.

Sample summary at S/N = 10, imaging polarimetry (PS1 g)

P = 10%: 0 targets within 1 h, 0 within 2 h, 0 within 4 h
P = 1%: 0 targets within 2 h, 0 within 4 h, 0 within 8 h

Caveats: g magnitudes are observed-frame, not corrected for Galactic extinction; the 1/P rule is approximate (a 3-sigma detection criterion via $\sigma_P = \sqrt{2}/(S/N)$ scales as $1/P^2$) – confirm convention before the proposal; several candidates lie at $z < 0.5$ (W2M rows) – vet before targeting.

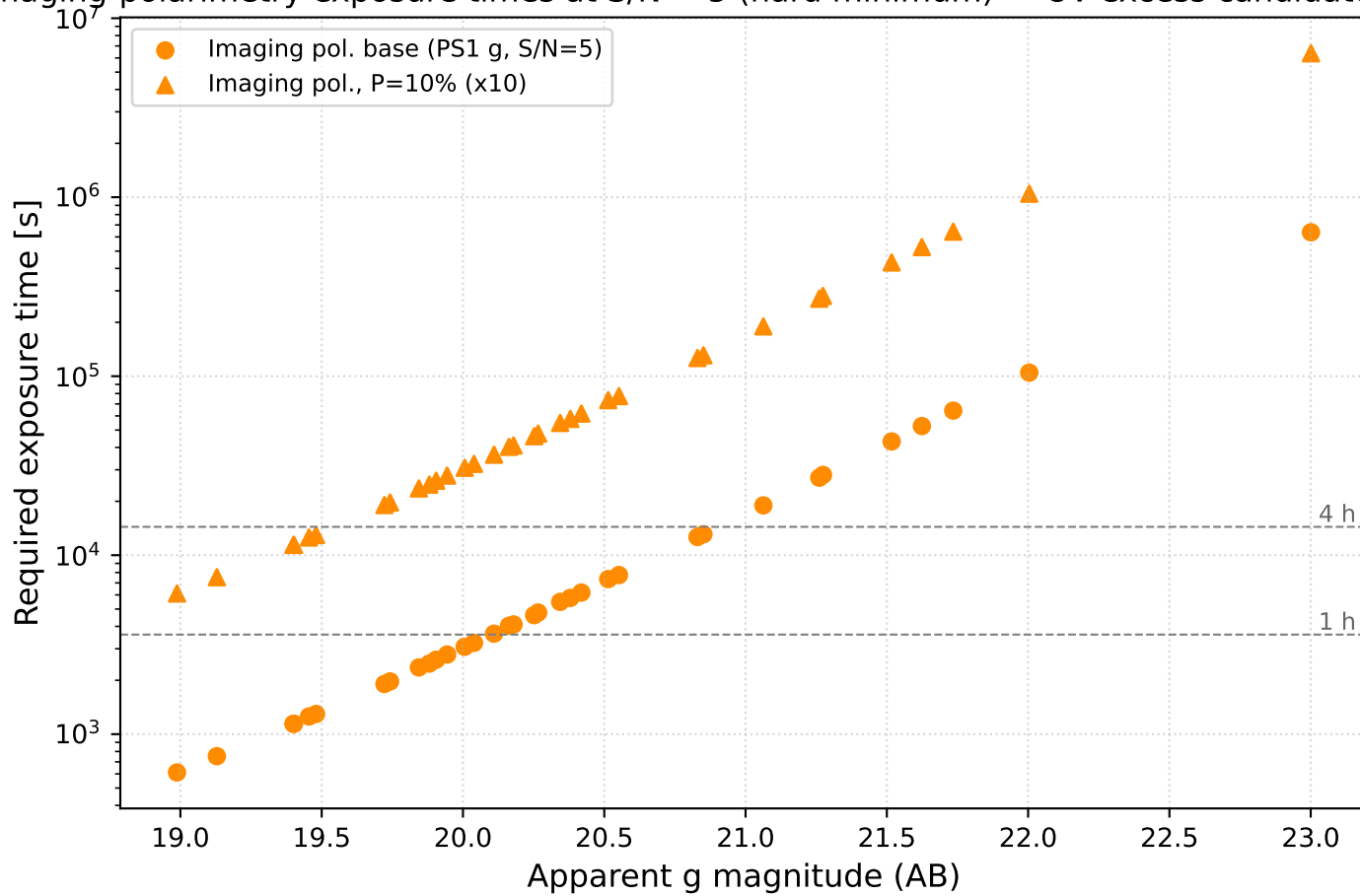
Kast imaging-polarimetry exposure times at S/N = 10 (ideal) — UV-excess candidates (34 targets)



Per-target imaging-polarimetry exposure times at S/N = 10 (ideal) (brightest first; PS1 g band)

Target	g (AB)	ETC mag	Img base	Img P=10%	Img P=1%	N exp (10%)
J204626.10+002337.6	18.99	19.74	39.2 m	6.5 h	65.4 h	14
J150746.11+093731.2	19.13	19.88	48.7 m	8.1 h	81.1 h	17
TARGETID_39628512285426154	19.40	20.15	74.5 m	12.4 h	124.1 h	25
TARGETID_39633384095352378	19.40	20.15	74.5 m	12.4 h	124.1 h	25
J150356.69+125914.3	19.45	20.21	82.0 m	13.7 h	136.6 h	28
J144809.53+054019.2	19.48	20.23	84.7 m	14.1 h	141.1 h	29
TARGETID_39628516710418961	19.72	20.47	2.1 h	20.9 h	208.8 h	42
J150357.14+041856.3	19.74	20.49	2.2 h	21.6 h	215.8 h	44
J141354.73+110047.5r	19.84	20.60	2.6 h	25.9 h	259.1 h	52
J152259.51+032347.6r	19.88	20.63	2.7 h	27.2 h	272.4 h	55
J143027.14+152514.8	19.91	20.66	2.9 h	28.6 h	286.4 h	58
J134039.67+051419.7	19.94	20.70	3.1 h	30.6 h	306.3 h	62
J150152.94+020604.9	20.01	20.76	3.4 h	33.9 h	338.9 h	68
J115449.31+112705.8	20.04	20.79	3.6 h	35.7 h	356.5 h	72
TARGETID_39633426973720698	20.11	20.86	4.0 h	40.1 h	401.5 h	81
J110018.80+080455.3	20.16	20.92	4.4 h	44.5 h	444.7 h	89
J132210.17+084232.9	20.18	20.93	4.5 h	45.2 h	452.4 h	91
TARGETID_39633267170738647	20.25	21.00	5.1 h	51.0 h	510.0 h	102
J150013.42+123645.2	20.27	21.02	5.3 h	52.8 h	527.8 h	106
TARGETID_39633136459448983	20.34	21.10	6.1 h	60.6 h	605.7 h	122
TARGETID_39633149587623462	20.38	21.13	6.4 h	63.8 h	638.0 h	128
TARGETID_39633325387681000	20.42	21.17	6.8 h	68.4 h	683.7 h	137
TARGETID_39633396137198490	20.51	21.27	8.1 h	81.3 h	813.4 h	163
TARGETID_39633432401152732	20.55	21.30	8.6 h	85.7 h	857.0 h	172
TARGETID_39633325958105931	20.83	21.58	14.0 h	140.1 h	1400.7 h	281
TARGETID_39633162619324602	20.85	21.60	14.5 h	145.1 h	1451.0 h	291
TARGETID_39633387157195822	21.06	21.81	21.1 h	210.5 h	2105.3 h	422
TARGETID_39632946793025886	21.26	22.01	30.1 h	300.8 h	3007.8 h	602
TARGETID_39633311366122701	21.27	22.03	31.2 h	311.7 h	3117.4 h	624
J140658.28+161142.6	21.52	22.27	48.0 h	479.6 h	4795.9 h	960
TARGETID_39628379145637914	21.62	22.38	58.5 h	584.7 h	5847.1 h	1170
TARGETID_39628346081936841	21.73	22.49	71.3 h	713.2 h	7131.8 h	1427
TARGETID_39633085267968176	22.00	22.76	116.3 h	1163.0 h	11629.7 h	2326
J143404.58+093528.5	23.00	23.75	707.6 h	7076.3 h	70762.8 h	14153

Fast imaging-polarimetry exposure times at $S/N = 5$ (hard minimum) — UV-excess candidates (34 targets)



Per-target imaging-polarimetry exposure times at S/N = 5 (hard minimum) (brightest first; PS1 g band)

Target	g (AB)	ETC mag	Img base	Img P=10%	Img P=1%	N exp (10%)
J204626.10+002337.6	18.99	19.74	10.2 m	1.7 h	17.0 h	4
J150746.11+093731.2	19.13	19.88	12.6 m	2.1 h	20.9 h	5
TARGETID_39628512285426154	19.40	20.15	19.0 m	3.2 h	31.7 h	7
TARGETID_39633384095352378	19.40	20.15	19.0 m	3.2 h	31.7 h	7
J150356.69+125914.3	19.45	20.21	20.9 m	3.5 h	34.9 h	7
J144809.53+054019.2	19.48	20.23	21.6 m	3.6 h	36.0 h	8
TARGETID_39628516710418961	19.72	20.47	31.8 m	5.3 h	53.0 h	11
J150357.14+041856.3	19.74	20.49	32.8 m	5.5 h	54.7 h	11
J141354.73+110047.5r	19.84	20.60	39.3 m	6.6 h	65.6 h	14
J152259.51+032347.6r	19.88	20.63	41.3 m	6.9 h	68.9 h	14
J143027.14+152514.8	19.91	20.66	43.4 m	7.2 h	72.4 h	15
J134039.67+051419.7	19.94	20.70	46.4 m	7.7 h	77.4 h	16
J150152.94+020604.9	20.01	20.76	51.3 m	8.6 h	85.5 h	18
J115449.31+112705.8	20.04	20.79	54.0 m	9.0 h	89.9 h	18
TARGETID_39633426973720698	20.11	20.86	60.7 m	10.1 h	101.2 h	21
J110018.80+080455.3	20.16	20.92	67.2 m	11.2 h	112.0 h	23
J132210.17+084232.9	20.18	20.93	68.4 m	11.4 h	113.9 h	23
TARGETID_39633267170738647	20.25	21.00	77.0 m	12.8 h	128.3 h	26
J150013.42+123645.2	20.27	21.02	79.7 m	13.3 h	132.8 h	27
TARGETID_39633136459448983	20.34	21.10	1.5 h	15.2 h	152.3 h	31
TARGETID_39633149587623462	20.38	21.13	1.6 h	16.0 h	160.3 h	33
TARGETID_39633325387681000	20.42	21.17	1.7 h	17.2 h	171.8 h	35
TARGETID_39633396137198490	20.51	21.27	2.0 h	20.4 h	204.2 h	41
TARGETID_39633432401152732	20.55	21.30	2.2 h	21.5 h	215.1 h	44
TARGETID_39633325958105931	20.83	21.58	3.5 h	35.1 h	351.1 h	71
TARGETID_39633162619324602	20.85	21.60	3.6 h	36.4 h	363.7 h	73
TARGETID_39633387157195822	21.06	21.81	5.3 h	52.7 h	527.2 h	106
TARGETID_39632946793025886	21.26	22.01	7.5 h	75.3 h	752.9 h	151
TARGETID_39633311366122701	21.27	22.03	7.8 h	78.0 h	780.3 h	157
J140658.28+161142.6	21.52	22.27	12.0 h	120.0 h	1199.9 h	240
TARGETID_39628379145637914	21.62	22.38	14.6 h	146.3 h	1462.7 h	293
TARGETID_39628346081936841	21.73	22.49	17.8 h	178.4 h	1783.9 h	357
TARGETID_39633085267968176	22.00	22.76	29.1 h	290.8 h	2908.4 h	582
J143404.58+093528.5	23.00	23.75	176.9 h	1769.2 h	17691.7 h	3539